Process Mining and Management 2025-2026

10 December 2025

Instructions

- Write clearly; illegible answers will not be marked.
- Take care to identify each answer clearly with:
  - the number of the exercise.
  - where appropriate, the part of the exercise you are answering.
- Clearly cross out unwanted answers before handing in your answers.
- You are not allowed to use lecture material.

BPMN Modeling and Analysis

Exercise 1 (BPMN Modeling)**.**

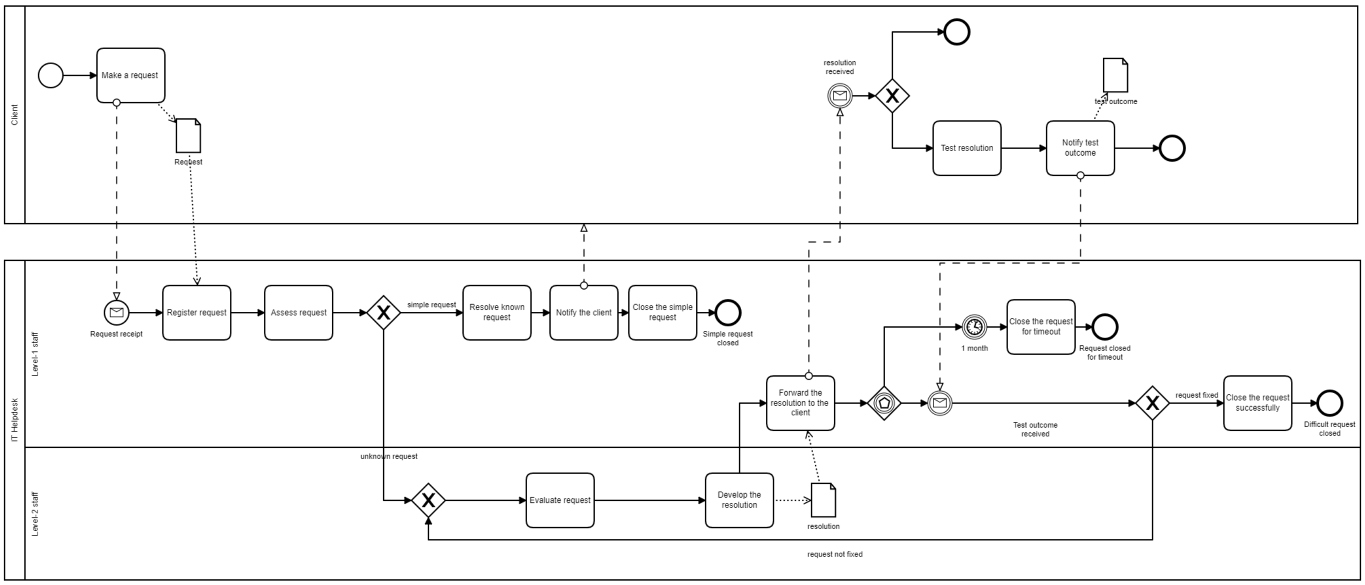
Let us consider the following scenario for client request handling at a IT helpdesk:

IT helpdesk request handling.[1]
The clients of the IT helpdesk are employees of a company. A request may be an IT-related problem that a client has, or an access request (e.g., requesting rights to access an IT system). Requests need to be handled according to their type and their priority.
A client makes a request to the help desk. The help desk staffed with "Level-1" support staff registers the request. Level-1 support staff typically are junior people with less than 12 months experience, but are capable of resolving simple requests. When the Level-1 employee knows how to resolve a request, she resolves it, notifies the resolution to the client and closes the simple request.
When the Level-1 employee does not know how to resolve a request, the request is forwarded to a more experienced "Level-2" support staff. When a Level-2 employee receives a request, the request is evaluated, a resolution is developed and sent back to the Level-1 employee. Eventually, the Level-1 employee forwards the resolution to the client who can test the resolution and notify the outcome to the IT helpdesk. The client notifies the outcome of the test to the Level-1 employee via e-mail. If the client states that the request is fixed, the request is closed successfully, and the process ends. If the request is not fixed, it is resent to Level-2 support for further action and goes through the process again. If, instead, after the client notification, no test outcome is received back for a month, the Level-1 employee closes the request for timeout and the process ends.

[1] The described scenario is inspired from a scenario reported in the *Handbook of Business Process Management, Vol. 1.*

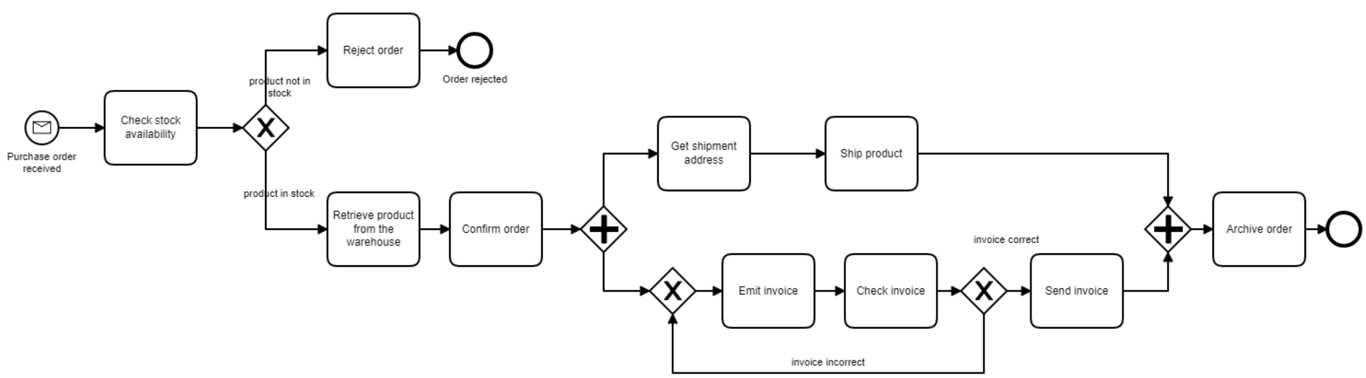
Model the scenario as a BPMN process model.

Answer. Note: this is one of the possible solutions



Exercise 2 (BPMN Understanding and analysis)**.**

Let us consider the process model reported below:



and the cycle and processing time of the activities reported in the table below:

Activity	Cycle time	Processing time
Check stock availability	1 day	30 min
Reject order	7 days	1 hour
Retrieve product from the warehouse	3 days	4 hours
Confirm order	4 days	2 hours
Get shipment address	1 day	30 min
Ship product	10 days	24 hours
Emit invoice	1 day	1 hour
Check invoice	1 day	30 min
Send invoice	2 days	30 min
Archive order	4 days	1 hour

1. Describe the process trying to clearly define the BPMN constructs used for modeling details on the control flow, resources and data objects.

2. Classify the activities in this process into three categories: value adding (VA), business value-adding (BVA) and non-value-adding (NVA).

3. Assuming that in 20% of the cases the orders are rejected (because of the lack of the product in the warehouse) and that in 10% of the cases there is an issue with the invoice and that a day corresponds to a working day of 8 hours, compute cycle time (CT), processing time (PT) and cycle time efficiency (CTE) for this process.

Answer.

1. The process model describes a purchase order process. Once the purchase order is received, it has to be checked against the stock to determine if the requested item(s) are available. Depending on stock availability, the purchase order is rejected and the process ended, or the product can be retrieved from the warehouse and the order confirmed. In the latter case, while the shipment address is retrieved and the product shipped, the invoice is emitted and checked. If the invoice is correct, it is sent, otherwise the invoice is emitted again and the invoice management iterated. The process completes by archiving the order.

Activity	Activity value
Check stock availability	BVA
Reject order	BVA
Retrieve product from the warehouse	BVA
Confirm order	VA
Get shipment address	BVA
Ship product	VA
Emit invoice	BVA (first time)
Check invoice	BVA (first time)
Send invoice	BVA
Archive order	BVA

2.

3.

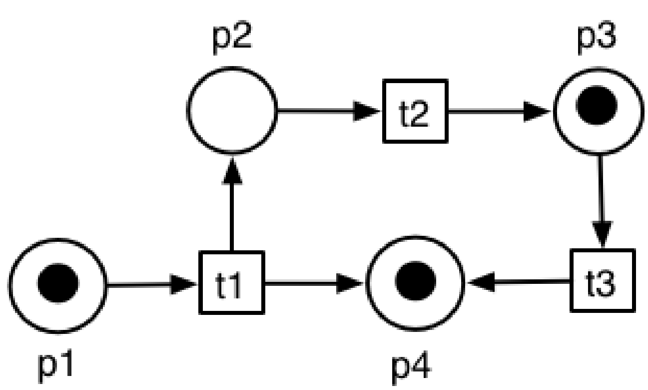
$$
\begin{aligned}
CT &= 1 + (0.2 * 7) + (0.8 * (3 + 4 + max(1 + 10, 2/0.9 + 2) + 4)) \\
&= 1 + 1.4 + (0.8 * (7 + 11 + 4)) \\
&= 2.4 * (0.8 * 22) \\
&= 2.4 + 17.6 \\
&= 20 \text{ days}
\end{aligned}
$$

$$
\begin{aligned}
PT &= 0.5 + (0.2 * 1) + (0.8 * (4 + 2 + max(0.5 + 24, 1.5/0.9 + 1) + 0.5)) \\
&= 0.5 + 0.2 + (0.8 * (6 + 24.5 + 1) + 0.5) \\
&= 0.7 * (0.8 * 31.5) \\
&= 0.7 + 25.2 \\
&= 25.9 \text{ hours}
\end{aligned}
$$

$$
\begin{aligned}
CTE &= 25.9 \text{ hours}/20 \text{ days} \\
&= 25.9 \text{ hours}/160 \text{ hours} \\
&= 16\%
\end{aligned}
$$

Petri Nets

Exercise 3 (Basic Petri Nets)**.**
Consider the Petri Net system depicted below.



1. Formalise the net as a 4-tuple $\langle P, T, F, m_0 \rangle$.
2. Give the preset and the postset of each transition.
3. Which transitions are enabled at m_0?
4. Give the list of all reachable markings by drawing the reachability graph corresponding to the Petri Net.

5. What are the reachable terminal markings? How do you identify them in the Transition System?
6. Is there a reachable marking where we have a non-deterministic choice?

Answer.

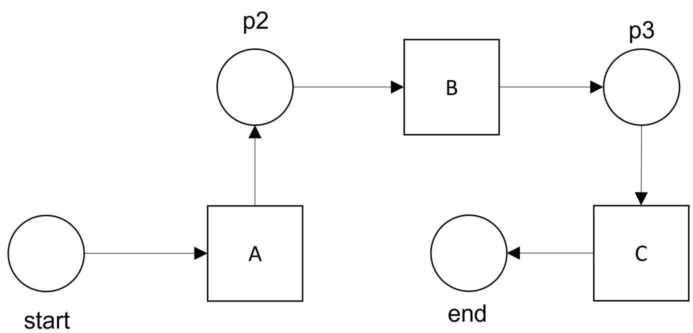
1. $P = \{p1, p2, p3, p4\}$, $T = \{t1, t2, t3\}$,
   $F = \{(p1, t1), (t1, p2), (p2, t2), (t2, p3), (p3, t3), (t3, p4), (t1, p4)\}$, $m_0 = [p1, p3, p4]$.
2. ${}^\bullet t1 = \{p1\}$, $t1^\bullet = \{p2, p4\}$, ${}^\bullet t2 = \{p2\}$, $t2^\bullet = \{p3\}$, ${}^\bullet t3 = \{p3\}$, $t3^\bullet = \{p4\}$.
3. $t1$ and $t3$;
4. Reachability graph

$$
\begin{tikzcd}
{[p1, p3, p4]} \arrow[r, "t3"] \arrow[d, "t1"] & {[p1, 2 \cdot p4]} \arrow[d, "t1"] \\
{[p2, p3, 2 \cdot p4]} \arrow[r, "t3"] \arrow[d, "t2"] & {[p2, 3 \cdot p4]} \arrow[d, "t2"] \\
{[2 \cdot p3, 2 \cdot p4]} \arrow[r, "t3"] & {[p3, 3 \cdot p4]} \arrow[d, "t3"] \\
 & {[4 \cdot p4]}
\end{tikzcd}
$$

   $m0 = [p1, p3, p4]$, $m1 = [p1, 2 \cdot p4]$, $m2 = [p2, p3, 2 \cdot p4]$, $m3 = [p2, 3 \cdot p4]$, $m4 = [2 \cdot p3, 2 \cdot p4]$, $m5 = [p3, 3 \cdot p4]$, $m6 = [4 \cdot p4]$.

5. $m6$
6. Yes. At markings $m0$ (where both $t1$ and $t3$ are enabled) and $m2$ (where both $t2$ and $t3$ are enabled).

Exercise 4 (Conformance checking)**.**
Consider the Petri Net system depicted below.



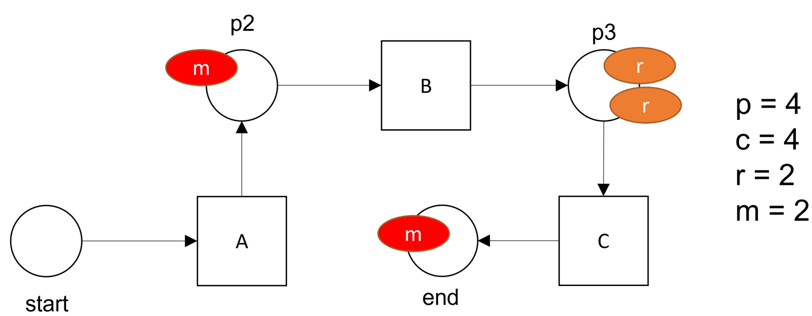
And trace $\langle A, B, B\rangle$. Replay the trace to check for conformance and

- Say whether it is conformant or not;
- Provide the numbers for p = produced tokens, c = consumed tokens, m = missing tokens, r = remaining tokens. In case of remaining or missing tokens say which are the problematic places (or draw the annotated Petri Net).

[Note: this is a simple net. The exercises in the exam may be slightly more complicated. Review the ones done during the lectures.]

Answer.

- The trace is not conformant.
- The values of p, c, m, and r are depicted below together with the positions where tokens are missing or remaining. remember that you have to increase the counter for consumed tokens at the end also if the token(s) do not reach the sink place.



Process Discovery

Exercise 5 (Heuristic Miner).

Let us consider the following event log extracted from some information system:

$$L = [\langle a, b, d\rangle^{100},$$
$$\langle a, c, b, d\rangle^{80},$$
$$\langle a, b, c, d\rangle^{2},$$
$$\langle a, c, d\rangle^{40},$$
$$\langle a, d\rangle^{1}]$$

1. Compute the frequency matrix $| >_L |$.
2. Compute the dependency measures matrix $| \Longrightarrow_L |$.
3. Construct the dependency graph using a threshold of 0.90 for $| \Longrightarrow_L |$, the threshold for direct succession ($| >_L |$) is 1.

Answer.

1. The frequency matrix ($| >_L |$)

	a	**b**	**c**	**d**
a	0	102	120	1
b	0	0	2	180
c	0	80	0	42
d	0	0	0	0

2. The dependency measure matrix ($| \Longrightarrow_L |$):

	a	**b**	**c**	**d**
a	0	0.99	0.99	0.5
b	-0.99	0	-0.94	0.99
c	-0.99	0.94	0	0.98
d	-0.5	-0.99	-0.98	0

3. The dependency graph